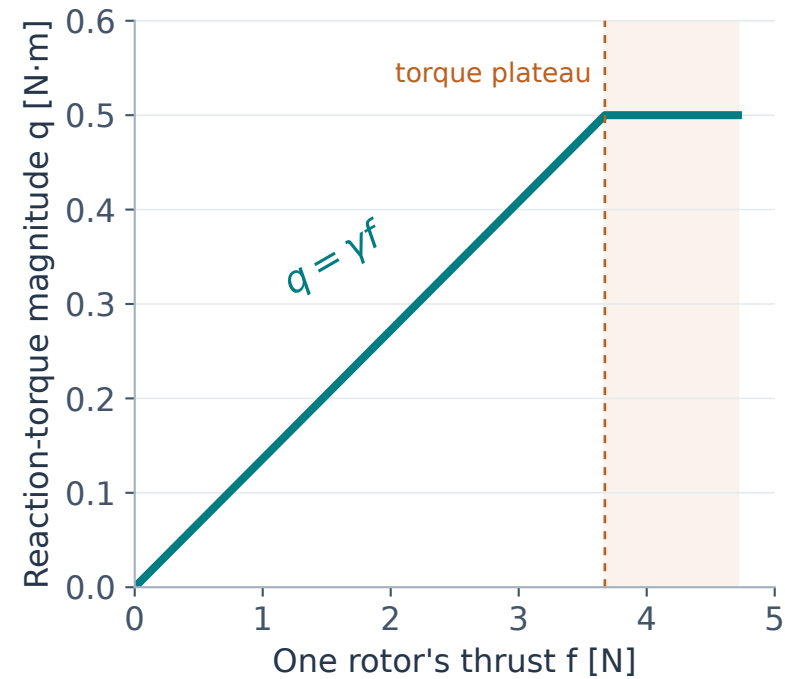


# Preserve the moments. Reduce collective when required.

At high thrust, the reaction-torque cap can eliminate yaw authority.

1 Each rotor has a torque cap

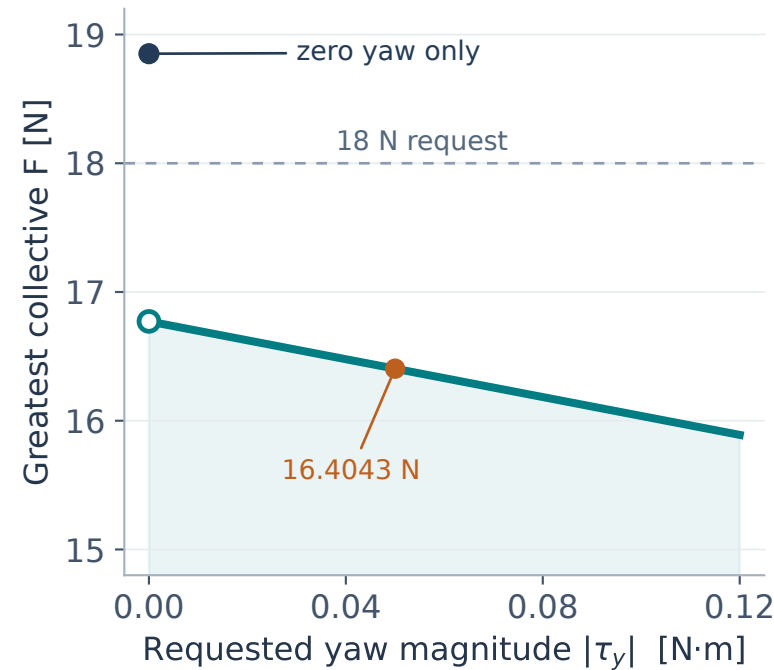


The cap starts at  $f_c = 3.6731$  N.  
More thrust then adds no reaction torque.

**MODEL**      $q_i = \min(\gamma f_i, c)$       $\tau_y = \sum_i d_i q_i$       $F = \sum_i f_i$       $0 \leq f_i \leq U$   
 $c = 0.5 \text{ N}\cdot\text{m}$       $U = 4.7127361 \text{ N}$       $\gamma = k_q/k_t = 0.13612565 \text{ m}$       $d = (-1, +1, +1, -1)$

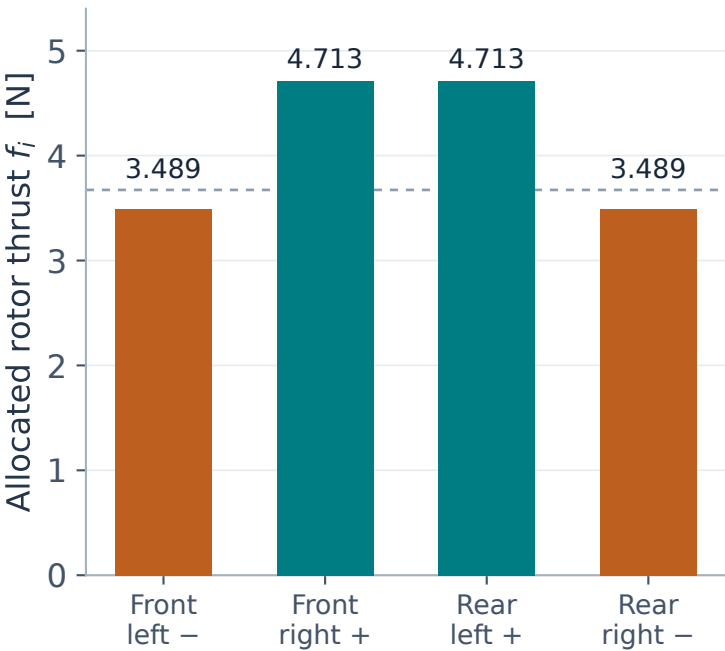
**SCOPE**     Analytical illustration of the explicit one-drone profile. These are steady forces, not measured motor transients.  
Implementation tolerances:  $10^{-12}$  N and  $10^{-12}$  N·m. Tiny requests can differ from the exact-model curve. No stability proof.

2 Yaw can lower the thrust ceiling



Zero roll and pitch moments; exact model.  
The jump comes from the torque plateau.

3 A feasible worked example



Request: 18 N and yaw +0.05 N·m.  
Result: 16.4043 N; roll/pitch moments zero.